

MuGate

Full-Project Soft / Application Relational Schema

Capstone-v2 · Declared FKs + application soft keys · PostgreSQL (Supabase)

Legend

- Solid line = declared PostgreSQL FOREIGN KEY
- - - - - Dashed line (soft) = application / soft key — column exists, NO DB FK

Soft relationships included (6)

- **CapstonePartners.userId -> Users.id**
JWT userId stored as text; ownership checks in CapstoneRepository
- **InternshipReviews.userId -> Users.id**
JWT userId stored as text; review CRUD scoped by userId
- **UserRoadmap.userId -> Users.id**
JWT userId stored as text; RoadMapRepository filters by userId
- **Admins.universityId -> Users.universityId**
Admin middleware + auth.routes LEFT JOIN on universityId
- **ScrapeQueue.runId -> ScraperRuns.id**
KnowledgeRepository.enqueueUrls tags queue rows with run UUID
- **Events.createdBy -> Users.name**
Manual events store display name (not id); denormalized attribution

Document pages

1. Page 1 — Full Schema Overview (FK + Soft)
2. Page 2 — Auth / Users (+ Admins soft)
3. Page 3 — Academic (+ UserRoadmap soft)
4. Page 4 — MuChat
5. Page 5 — Knowledge / Scraper (+ runId soft)
6. Page 6 — Capstone, Internships & Events

Truly unrelated after soft links

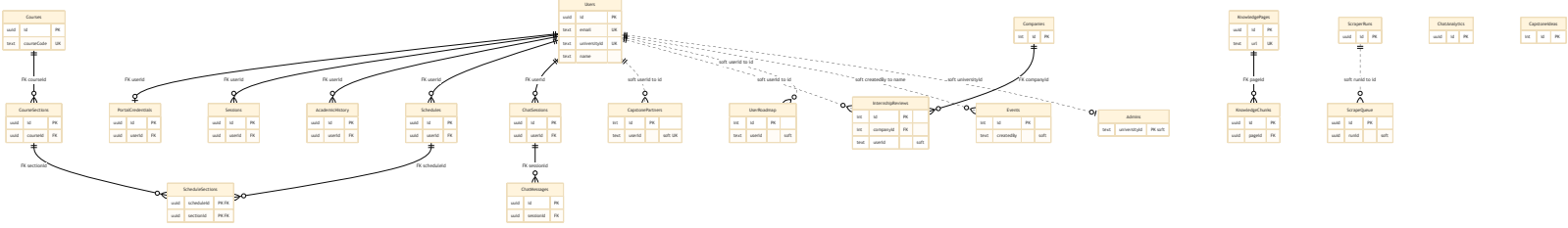
- ChatAnalytics - aggregate privacy-safe metrics only (no sessionId / userId columns)
- CapstoneIdeas - standalone idea catalog (no owner / user columns)

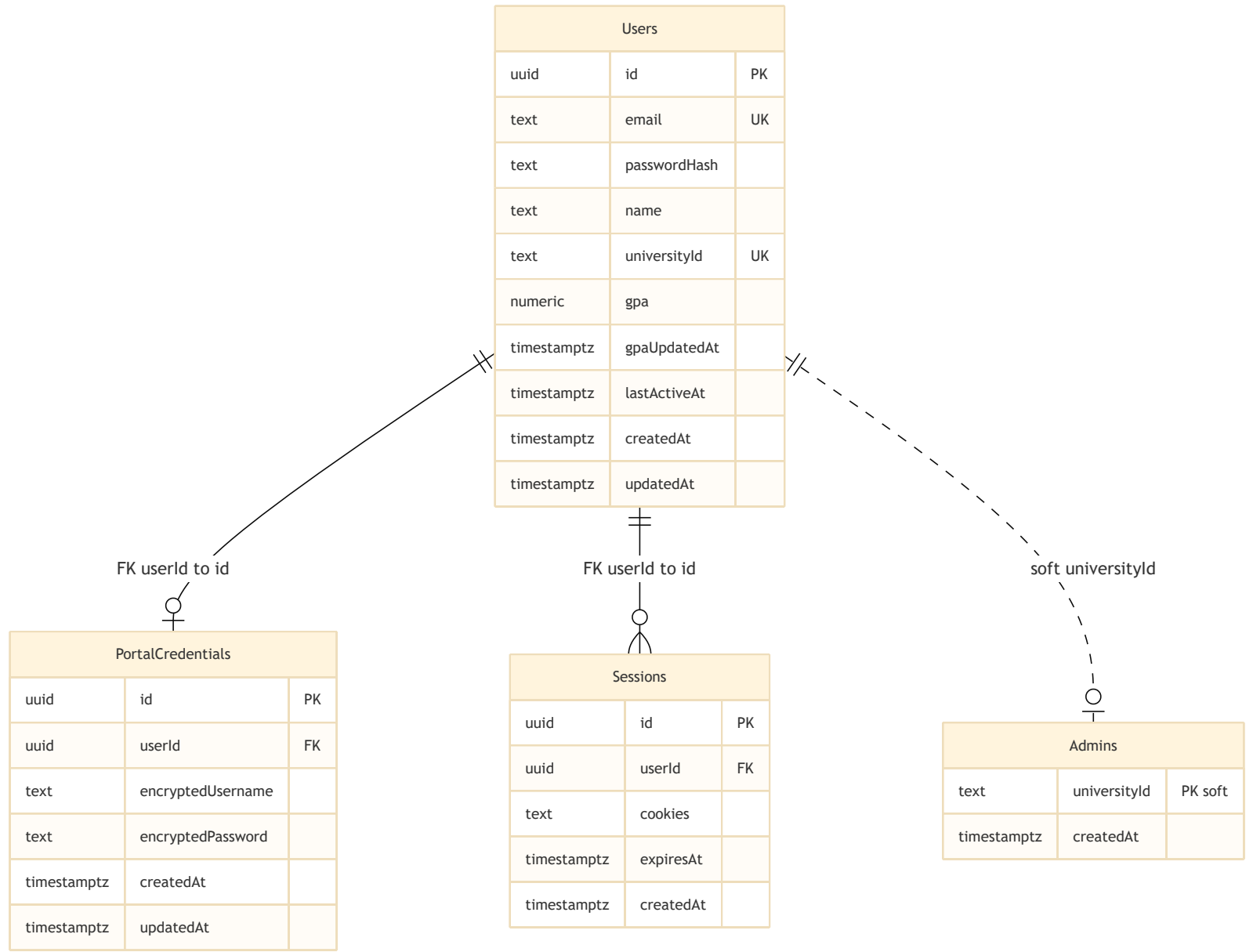
Why soft keys exist

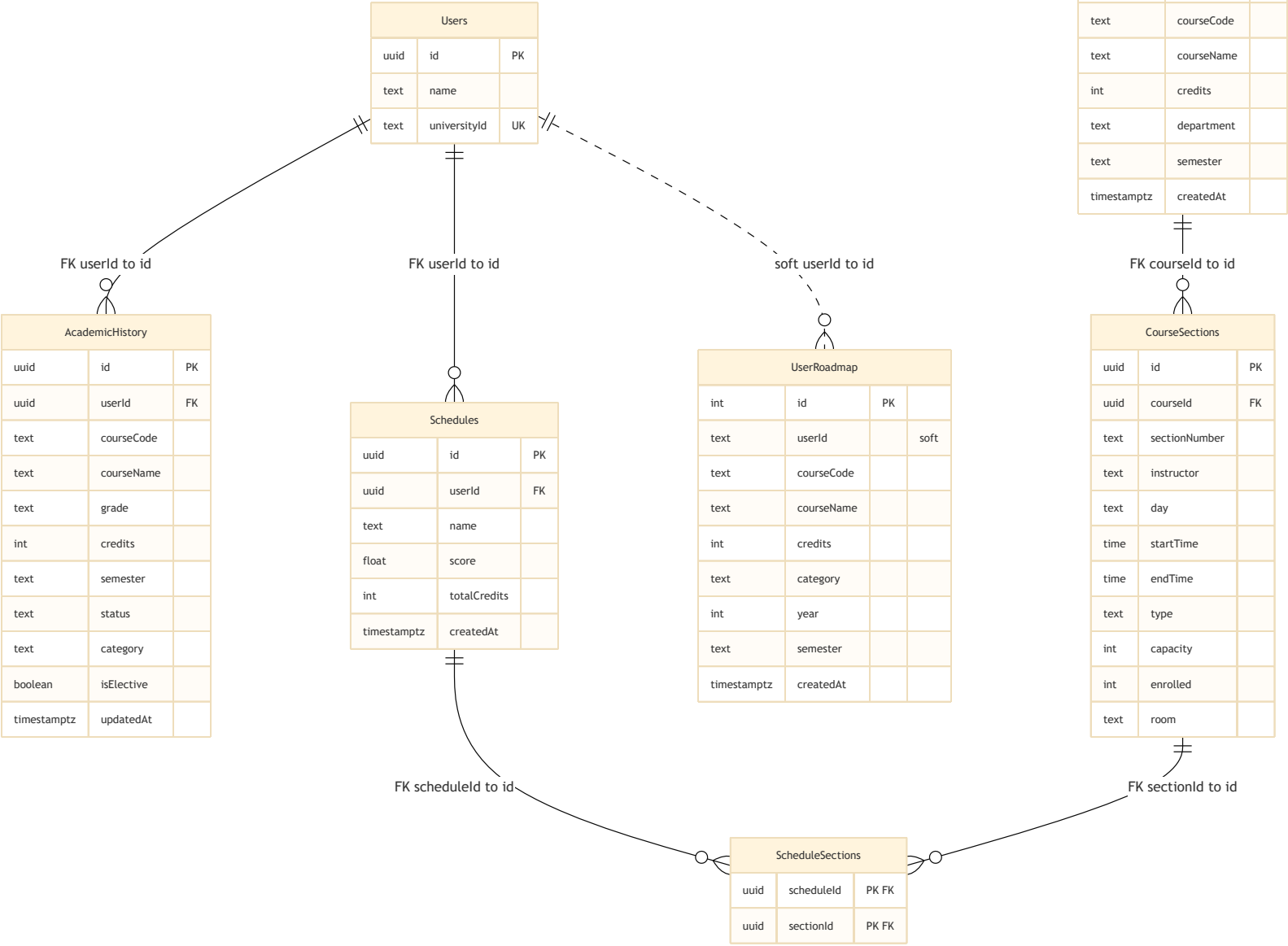
- Legacy SQL Server ensureTables used NVARCHAR userId (text) without FK declarations.
- UUID user ids stored as text in feature tables → type mismatch blocks easy PG FKs.
- Admins keyed by portal universityId (natural key), not Users.id UUID.
- ScrapeQueue.runId left nullable / unconstrained so queue rows survive run cleanup.
- Events.createdBy is an audit display string (name), not a resolvable user UUID.

Derived from Capstone-v2 / MuGate PostgreSQL schema (supabase/migrations/20260714120000_mugate_schema.sql) plus verified soft keys from backend repositories/services/routes. Strict FK-only companion: MuGate-Full-Project-Relational-Schema.pdf

Not drawn as soft FKs: AcademicHistory.courseCode / UserRoadmap.courseCode -> Courses (shared natural key text, but no app join). Events.sourceId is external scraper id.







Users		
uuid	id	PK
text	name	
text	email	UK



FK userId to id



ChatSessions		
uuid	id	PK
uuid	userId	FK
text	title	
text	source	
boolean	isPinned	
boolean	isActive	
timestampz	createdAt	
timestampz	updatedAt	



FK sessionId to id



ChatMessages		
uuid	id	PK
uuid	sessionId	FK
text	role	
text	content	
int	tokensUsed	
timestampz	createdAt	

ChatAnalytics		
uuid	id	PK
text	questionCategory	
boolean	isFailed	
int	responseTimeMs	
timestampz	createdAt	

KnowledgePages		
uuid	id	PK
text	url	UK
text	title	
text	content	
text	contentHash	
text	category	
text	subcategory	
text	language	
int	wordCount	
text	sourceDomain	
timestamptz	lastScrapedAt	
boolean	isActive	
timestamptz	createdAt	
timestamptz	updatedAt	

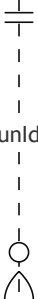
FK pageId to id



KnowledgeChunks		
uuid	id	PK
uuid	pageId	FK
int	chunkIndex	
text	content	
text	keywords	
text	category	
text	title	
text	entityType	
timestamptz	chromaSyncedAt	
text	embeddingModel	
timestamptz	createdAt	

ScraperRuns		
uuid	id	PK
text	runType	
text	status	
text	baseUrl	
int	pagesScraped	
int	pagesUpdated	
int	pagesNew	
int	pagesUnchanged	
int	errorCount	
text	errorDetails	
timestamptz	startedAt	
timestamptz	completedAt	

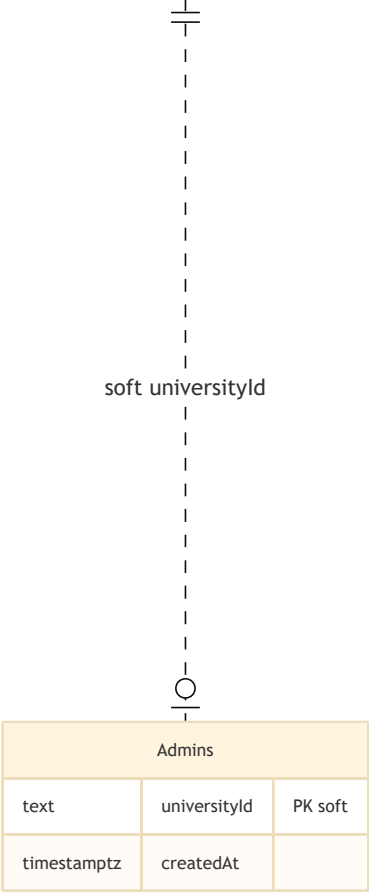
soft runId to id



ScrapeQueue			
uuid	id	PK	
text	url		
int	priority		
text	status		
uuid	runId		soft
int	depth		
timestamptz	createdAt		
timestamptz	updatedAt		

Users		
uuid	id	PK
text	universityId	UK
text	name	

soft universityId



Admins		
text	universityId	PK soft
timestamptz	createdAt	

Users		
uuid	id	PK
text	name	
text	email	UK
text	universityId	UK

Companies		
int	id	PK
text	name	
text	description	
text	colors	
float	scale	
text	svgString	
text	email	
text	phone	
text	website	
boolean	forceWhiteBack	
boolean	forceBlackBack	
boolean	isMetallic	

CapstoneIdeas		
int	id	PK
text	title	
text	description	
text	faculty	
int	year	
text	tags	
boolean	isActive	
timestampz	createdAt	

CapstonePartners			
int	id	PK	
text	userId		soft UK
text	userName		
text	email		
text	phone		
text	major		
text	skills		
text	description		
text	lookingFor		
timestampz	createdAt		

InternshipReviews			
int	id	PK	
int	companyId	FK	
text	userId		soft
text	userName		
int	rating		
text	feedback		
timestampz	createdAt		

Events			
int	id	PK	
text	title		
text	description		
text	location		
timestampz	startDate		
timestampz	endDate		
text	category		
text	tags		
text	imageUrl		
text	externalUrl		
text	source		
text	sourceId		
text	scraperSource		
text	organizer		
boolean	isFree		
boolean	isActive		
text	createdBy		soft
timestampz	createdAt		
timestampz	updatedAt		

soft userId to id

soft userId to id

FK companyId to id

soft createdBy to name